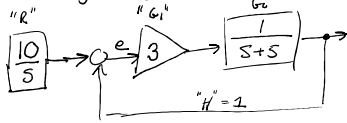


Calculating Steady State error



$$e(s) = \lim_{s \rightarrow 0} s \left\{ \left(1 - \frac{G_c G_p}{1 + G_c G_p H} \right) R \right\} = \lim_{s \rightarrow 0} s \left\{ \left(1 - \frac{3 \cdot \frac{1}{s+5}}{1 + 3 \cdot \frac{1}{s+5}} \right) \frac{10}{s} \right\}$$

$$= \lim_{s \rightarrow 0} s \left\{ \left(1 - \frac{3}{s+5} \right) \frac{10}{s} \right\} = \left(1 - \frac{3}{5} \right) 10 = 6.25 \quad \uparrow P1$$

Control design tools



$$T(s) = \frac{KG}{1+KGH}$$

Poles: $1+KGH=0$ C.E.

Solve for all values of K
Z Plot $\rightarrow$ RL

$$1+KGH=0 \quad KGH = -1 = 1 \angle 180^\circ$$



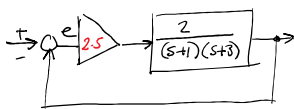
Magnitude Condition $|KGH| = 1$

Angle Condition $\angle KGH = 180^\circ$

$\Rightarrow$ Used for design
Used to sketch RL diagram

Example $G = \frac{2}{(s+1)(s+3)}$ design a Proportional controller for 4.33%OS

① draw the Block diagram



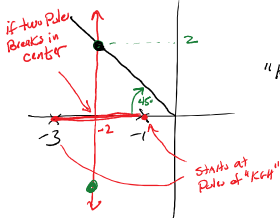
$$"KGH": \frac{K2}{(s+1)(s+3)}$$

$$M: \left| \frac{K2}{(s+1)(s+3)} \right| = 1$$

$$A: \angle \frac{K2}{(s+1)(s+3)} = 180^\circ$$

②A Find "design Point" (Requirements)

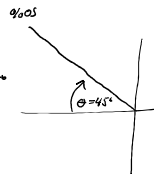
②B Sketch RL diagram



$$"KGH": \frac{K2}{(s+1)(s+3)}$$

$$\text{design Point: } s_{1,2} = -2 \pm 2j$$

$$\cos \phi = \zeta$$



$$\zeta = 0.707$$

$$\phi = \cos^{-1}(\zeta) = 45^\circ$$

③ Control design with mag/Angle Condition

$$M: \left| \frac{K2}{(s+1)(s+3)} \right| = 1 \quad \text{evaluate at the design Point}$$

$$@ s = -2 + 2j$$

$$K = \frac{(s+1)(s+3)}{2} = \frac{(-2+2j+1)(-2+2j+3)}{2} = \frac{(-1+2j)(1+2j)}{2}$$

$$= \frac{\sqrt{(-1)^2 + 2^2} \sqrt{1^2 + 2^2}}{2} = 2.5$$

$$\text{What is settling time? } T_s = \frac{4}{\sigma} = \frac{4}{2} = 2 \text{ seconds.} \quad \uparrow \#2$$

Example: Improve settling time by a factor of 2 with same %OS

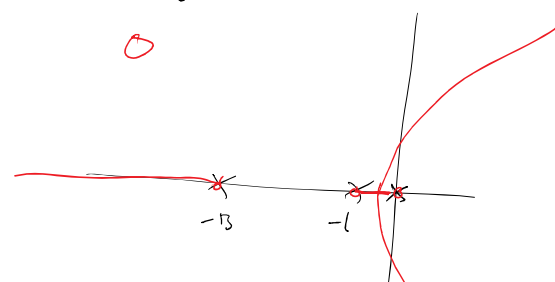
$$T_{s\text{new}} = \frac{T_{s\text{old}}}{2} = \frac{2}{2} = 1 \text{ second}$$

$$\sigma_{\text{new}} = \frac{4}{T_s} = \frac{4}{1} = 4$$

$$s_{1,2} = -4 \pm 4j$$

$\rightarrow$ Can not do with Proportional

try integral

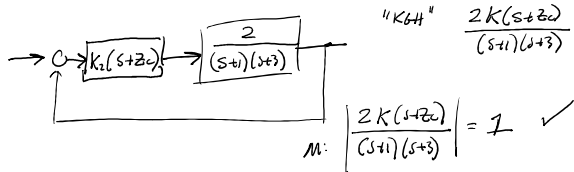


$$s_{12} = -4 \pm 4j$$

→ Can not do with Proportional Control!

Propose a derivative type compensator:

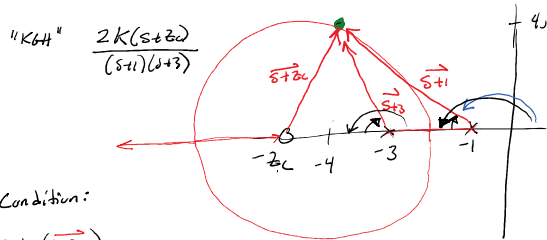
① draw Block diagram



②A $s_{12} = -4 \pm 4j$

②B Sketch RL

A: $\angle \frac{2K(s+2c)}{(s+1)(s+3)} = 180^\circ$



Angle Condition:

A: $\angle \frac{2K(s+2c)}{(s+1)(s+3)} = 180^\circ$

$\angle(s+1) = 180^\circ - \arctan\left(\frac{4}{3}\right) = 127^\circ$

$\angle(s+3) = 180^\circ - \arctan\left(\frac{4}{1}\right) = 104^\circ$

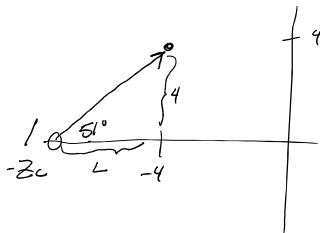
A: $\angle(s+2c) - \angle(s+1) - \angle(s+3) = 180^\circ$

$\angle(s+2c) = 180^\circ + 127^\circ + 104^\circ = 411^\circ - 360^\circ = 51^\circ$

trig:

$\tan 51^\circ = \frac{4}{L}$

$L = \frac{4}{\tan 51^\circ} = 3.2$



$2c = 4 + 3.2 = 7.2$

$K = 0.5$ May need → too low before

$\frac{K(1/s)}{s+1/\gamma} \rightarrow \frac{K/\gamma}{s+1/\gamma}$

